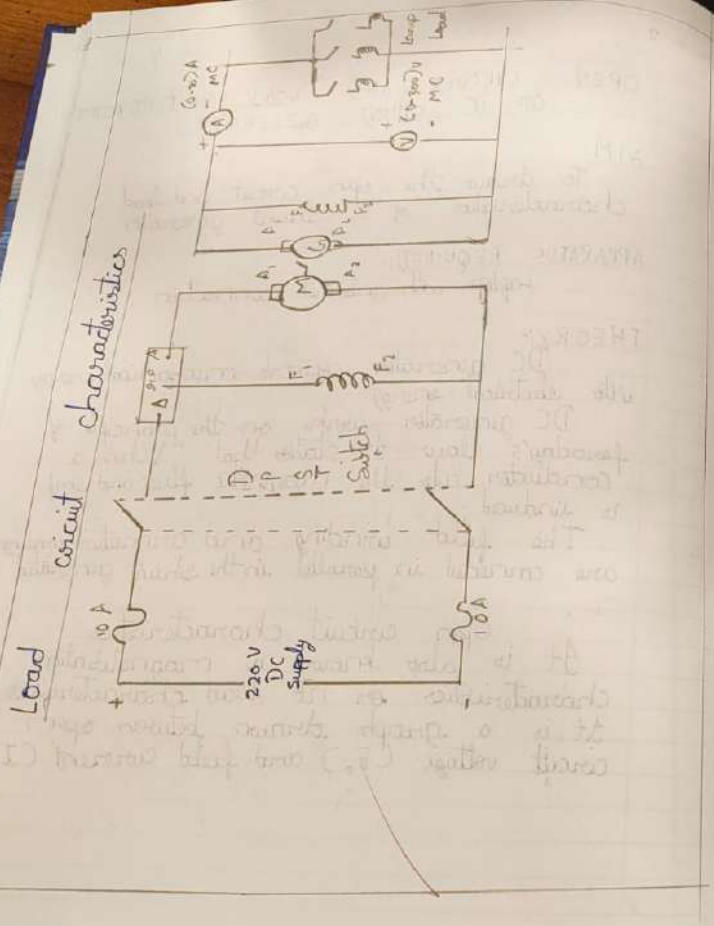


Expt. No.

Load circuit characteristic

It is a graph drawn between load voltage (V_L) and load circuit (I_L). It is also known as external characteristics.

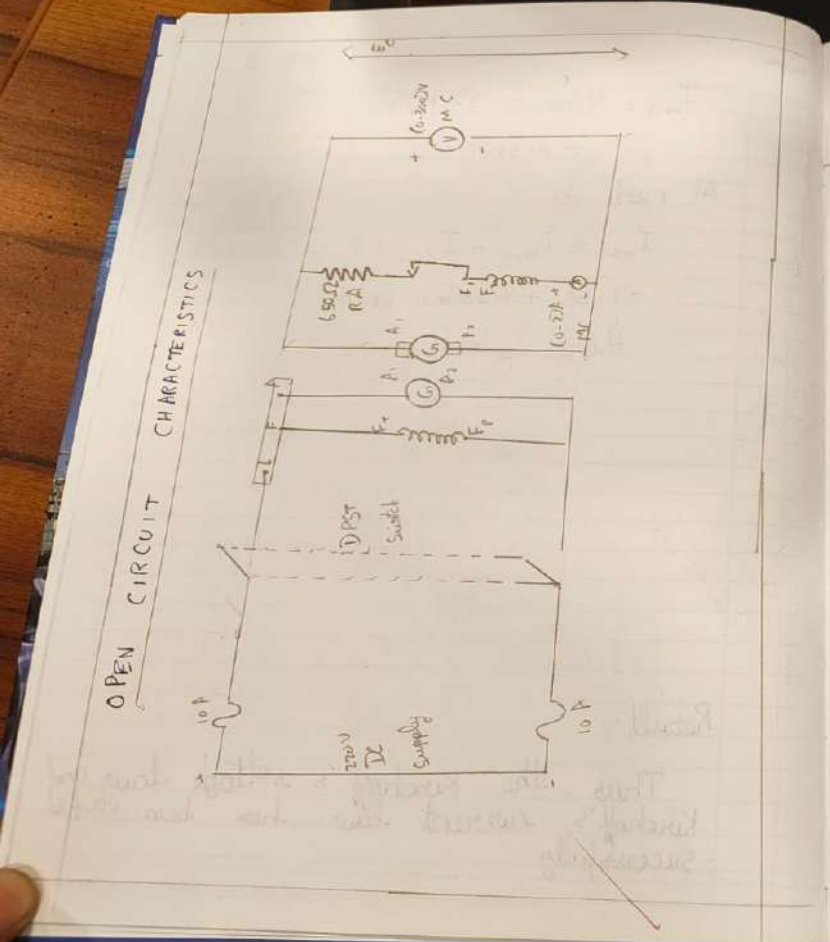
PROCEDURE:

Open circuit characteristic:

1) Connections are made as per the circuit diagram.
2) Switch on the supply and note down the readings. (Voltage and current)
3) By varying the field rheostat different values of open circuit voltage and field current are noted.
4) Plot the graph between open circuit voltage and field current.

Load circuit characteristic:-

1) Connections are made as per the circuit diagram.
2) Switch on the supply and note down the readings (for load voltage and current)
3) By varying and connecting different nodes, different values of load voltage and current.
4) Pick the graph between load voltage and load current.



Expt. No. 2

Page No. 4

OPEN CIRCUIT AND LOAD CHARACTERISTICS OF DC SHUNT GENERATOR

AIM:

To draw the open circuit and load characteristics of DC shunt generator

APPARATUS REQUIRED:-

Laptop with internet connection

THEORY:-

DC generator converts mechanical energy into electrical energy.

DC generator works on the principle of faraday's law. It states that when a conductor cuts the magnetic flux and emf is induced.

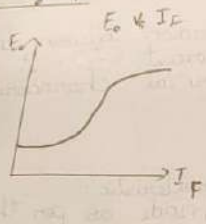
The field winding and armature windings are connected in parallel in the shunt generator

Open circuit characteristics

It is also known as magnetisation characteristics or no load characteristics. It is a graph drawn between open circuit voltage (E_0) and field current (I_f).

Open circuit Characteristics

Model Graph:



SNo	Voltage (V)	Current (A)
1	0	0
2	115	0.16
3	120	0.18
4	126	0.20
5	129	0.21
6	133	0.23

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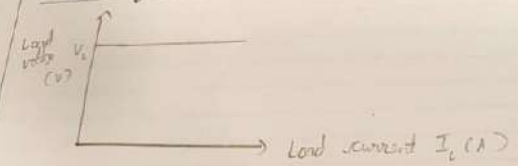
Load Characteristics

Model Graph:-



Observation table :-

S.No	Current (A)	Voltage (V)
1	1.2	220
2	2.8	212
3	3.2	208
4	3.6	205
5	5.5	200
6	7	195

Expt. No.

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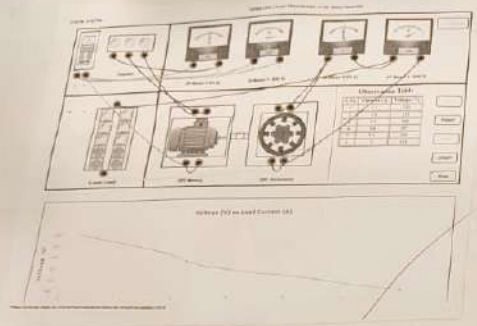

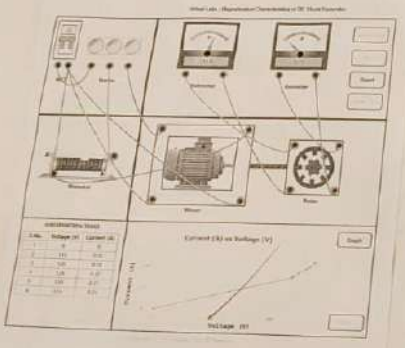

Expt. No.

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RESULT:-

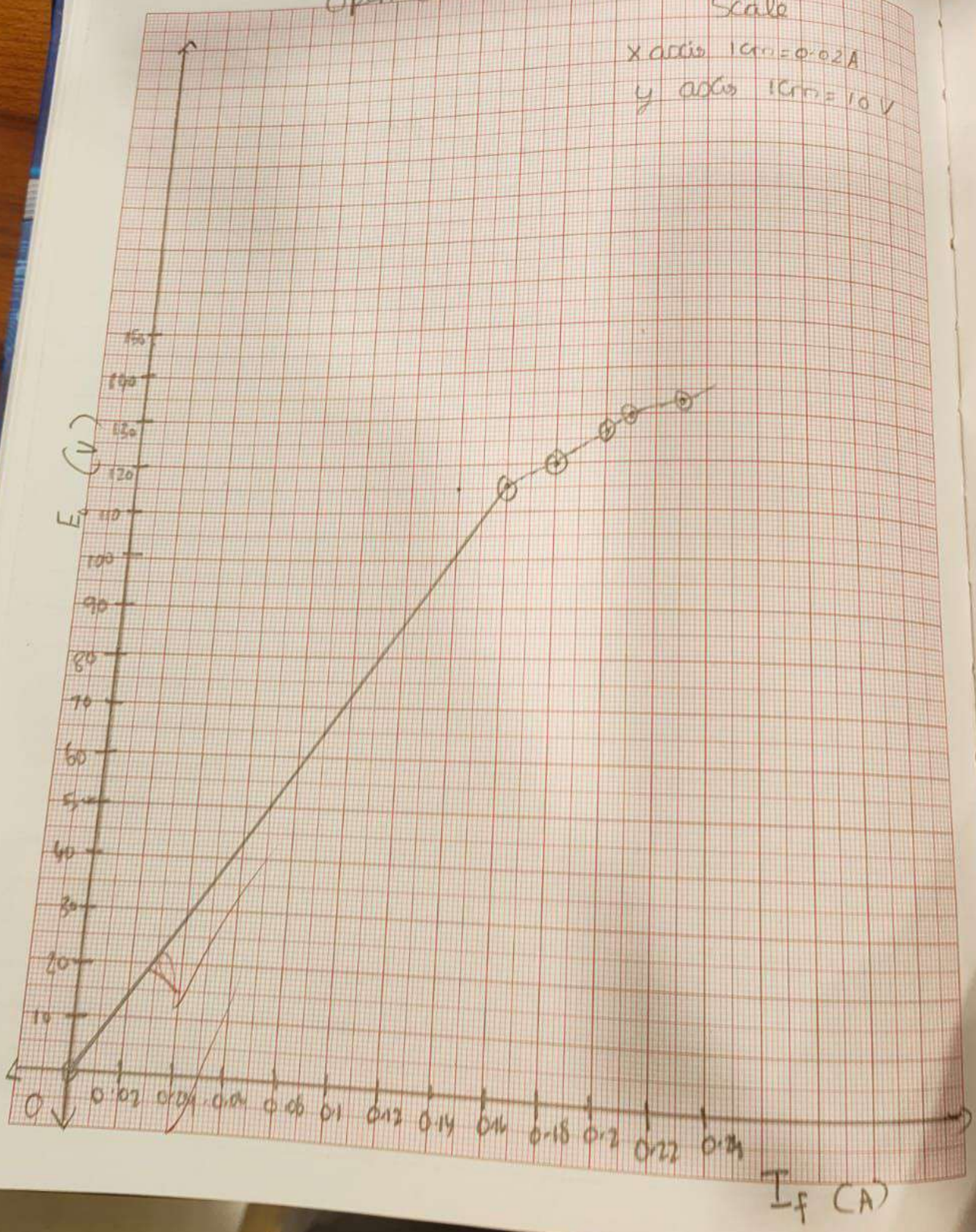
Thus, the open circuit and load characteristics of DC shunt generator were verified successfully.

Open circuit characteristic

Scale

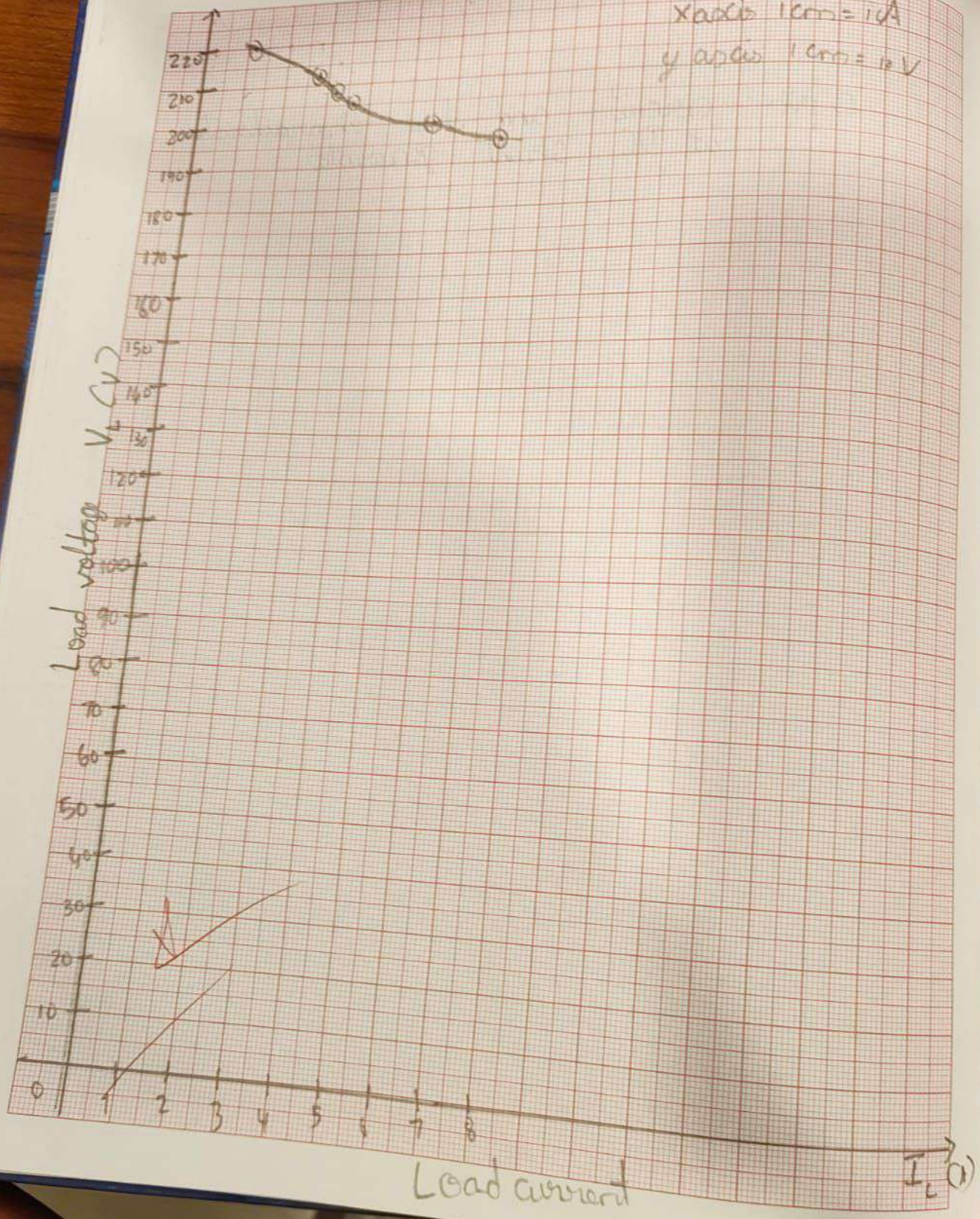
x axis 1cm = 0.02 A

y axis 1cm = 10 V



Load characteristic

Scale

X axis 1cm = 1A

y axis 1cm = 10 V